

A Quantitative Modeling of Protein-DNA interaction for Improved Energy Based Motif Finding Algorithm

Junguk Hur

School Of Informatics, Indiana University, Bloomington

Abstract

Finding motifs is an important challenge in computational biology. This paper describes an improved Energy-Based Motif Finding (EBMF) algorithm employing the chromatin immunoprecipitation on microarray (ChIP-on-Chip) experiments data. Unlike most other computational methods for motif discovery, based on the strong-signal model, the EBMF algorithm can utilize additional weak quantitative signal information from the ChIP-on-Chip data. Our ultimate goal is to improve the currently available EBMF algorithm by incorporating association information between positions in a motif and by allowing the length of target motif not to be fixed. Up to now, we have implemented basic features of the EBMF using QR-decomposition and Downhill-Simplex Method for solving linear equations to get appropriate energy matrix representing the given TF binding sites. The implemented EBMF algorithms was applied to the yeast ChIP-on-Chip data set with three transcription factors; GAL4, GCN4, and RAP1. Unlike our expectation, the implemented program could not correctly find the known TFBS motifs due to the singularity problems.

1 INTRODUCTION

To understand the regulation of gene expression, the process by which a DNA segment is decoded to form a protein, is one of the greatest challenges in the current molecular biology. Transcription and translation are two main steps for gene expression. During the transcription process, mRNA molecule is created by copying a gene segment from the DNA. In the translation process, the mRNA is decoded to produce a protein.

To initiate the transcription process for a gene, one or more corresponding proteins, called transcription factors, should bind to specific regions, usually called binding sites, in the promoter region of the

gene. Generally, it is known that a transcription factor can bind to multiple different binding sites, but these sites usually have similar DNA sequence pattern and length. For most transcription factors, the common patterns for their binding sites, called motifs, are still unknown.

Most of the currently available motif finding algorithms[2, 4, 7] are extremely based on the strong-signal model in which only strong-signal sequences (i.e. those that are already known to contain binding sites very similar to the motif) are considered as input and weak-signal sequences (i.e. those do not contain any sub-string very similar to the motif) are disregarded. Buhler and Tompa have studied the limitations of methods based on the

strong-signal model[4] and Francis *et al* have further characterized the types of problematic instances that cannot be solved by most of the motif finding methods by introducing the notion of minimal number of binding sites rather than minimal number of strong signal sequences[5].

A recent trend in motif-finding is to make use of computational methods based on microarray data allowing the genome-wide location analysis. These methods includes chromatin immunoprecipitation on Chip (ChIP-on-Chip) which can specifically detect protien-DNA interaction *in vivo* and Protein Binding Microarray (PBM) which has been designed to extract protein-DNA interaction information *in vitro*[3, 5, 9]. These technologies gives us information of how strongly a given transcriptional factor binds to a specific genomic region as well as where it binds to.

Francis *et al.* have introduced an energy-based motif finding model to capture previously-ignored weak-signal information[5]. They used additional information from ChIP-on-Chip experiments and considered the binding strength (as measured experimentally by observing color intensity) of each available sequence. It can be easily assumed that the binding strength should relate to the degree of similarity between the motif and the binding sites in each sequence.

Based on the binding strength, their model considered the amount of signals that a sequence actually contains. By comparison with the results of other famous motif finding programs such as AlignACE and MEME, they have shown the high performance of their Energy-Based Motif Finding (EBMF) algorithm. However, their algorithm does not consider any association between positions in each motif and the length of the expected motif should be fixed at the first step.

Our ultimate goal of this project is to develop an improved EBMF algorithm that can utilize association information between positions in a motif, whose length may not be known. For this project, we have two distinctive phases; the first is to implement the basic features of EBMF and the second is to improve the algorithm equipped with the features described above such as incorporating distantly related positional relations.

2 METHODOLOGY

2.1 Energy-Based Model

The following energy-based model is based primarily on the Francis *et al.*' previous work[5]. In the microarray experiment, some copies of transcription factor (TF) may bind to the binding sites of different sequences s_i in the genome while some copies of the transcription factor and sequences are free. Let e_i be the average binding energy between the transcription factor and sequence s_i . Then $e_i = \ln(K_e)$ where the binding constant $K_e = [TF \bullet s_i] / [TF][s_i]$ (number of sequences bound to the transcription factors over number of free sequences) with the binding reaction modeled by $TF + s_i \leftrightarrow TF \bullet s_i$. For each sequence, we have a color intensity which represents the value of K_e . Thus, the color intensity can be used to estimate the binding energy between the transcription factor and each input sequence.

Let sequence s_i contain B_i binding sites and $E(b_{ij}, M)$ be the binding energy between the transcription factor and the j -th binding site b_{ij} in sequence s_i . The probability that the transcription factor binds to b_{ij} is

$$P_{ij} = \frac{e^{-E(b_{ij}, M)}}{\sum_{k=1}^{B_i} e^{-E(b_{ik}, M)}} \quad (1)$$

and we define $E_{total}(s_i, M)$ to be the expected binding energy between the TF and sequence s_i when there are at least one binding site in s_i that is

bound by the TF.

Now, we want to find a $4 \times l$ energy matrix M to minimize the following *prediction error*, where l is the length of binding sites and t sequences $S = \{s_i\}$ in which each sequence s_i has a corresponding binding energy e_i

$$\sum_{i=1}^t (E_{total}(s_i, M) - e_i)^2 \quad (2)$$

The initial energy matrices are refined to minimize the prediction error in EM-like iteration. Until there is no further decrease in the prediction error or the number of iterations reaches a given value, the iteration process is repeated.

2.2 Solving Linear Equations

We formulate an equation by setting expected binding energy to the given binding energy of that sequence, that is, $E_{total}(s_i, M') = e_i$. We only need $4 \times l$ distinctive equations to obtain a new energy matrix. In addition to the QR decomposition method[11] that was employed by Francis *et al.*, we also implemented Downhill Simplex Method[8, 11] to solve the linear equations since the matrices to be decomposed are singular up to the computational precision in some cases.

To create the linear equations, we convert each character in b_{ij} for any j in BS_{ik} to a 4-dimensional vector by using (1,0,0,0), (0,1,0,0), (0,0,1,0), and (0,0,0,1) for "A", "C", "G", and "T" respectively. The resulting $4l$ -dimensional vector v_{ij} is used to represent the binding site b_{ij} of length l . Then, the equations are represented as follows,

$$\left\{ \sum_{j=1}^{B_i} \left[\frac{p_{ij}}{1 - \prod_{BS_{ik}} \left(1 - \sum_{b_{im} \in BS_{ik}} P_{im} \right)} \right] \times v_{ij} \right\} \times V(M')^T = e_i \quad (3)$$

Here BS_{ik} means the divided groups of B_i binding sites of b_{ij} . For any two binding sites in the same group BS_{ik} , the distance between them is within d_{min} while the distance between any two binding sites in different groups is larger than d_{min} .

2.3 Test Data

ChIP-on-Chip experiment data with yeast Gal4, Raf1, and Gcn4 transcription factors were obtained from the Harbison *et al.*'s paper[6]. The following Table 1 shows the four transcription factor and the consensus sequences of their respective DNA binding sites from TRANSFAC[1, 10].

TF	Length	Consensus (TRANSFAC)
GAL4	17	CGGN11CCG
RAP1	11	[A G][A T]ACCCATACA
GCN4	10	[A C]AATGACTCA

Table 1: Yeast transcription factors and their binding site consensus

3 IMPLEMENTATION

The first phase of this project that aims at implementing the basic features of EBMF has been undertaken. The basic features of EBMF with QR-decomposition and Downhill Simplex Method have been implemented in Perl. The program can perform iterative searches of candidate motif matrix from a given initial matrix.

4 RESULTS

A test run on the yeast GAL4, GCN4, and RAP1 motif data showed that our current program could not predict proper energy matrix even though we have proper initial energy matrix. The Figure 2

shows the TFBS motifs (in the forms of energy matrix) found by our EBMF programs. Though these matrixes had minimal energy differences from the original binding energies, they were still far deviated from the expected and previously known TFBS motifs in the TRANSFAC. The GCN4 has been found relatively close to the known motif but the other two transcription factors of GAL4 and RAP1 showed completely inconsistent results.

a) GAL4

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
A	0.17	1.00	1.17	-0.50	-3.24	0.27	-0.50	-0.50	1.20	0.33	-1.64	-0.68	-1.48	0.61	0.77	0.31	-0.09
C	-0.44	2.21	1.01	-0.66	-2.89	0.45	0.97	0.22	1.23	-0.60	-1.54	-0.07	-0.94	0.88	0.29	0.90	1.24
G	-1.86	1.66	0.60	-0.06	-3.05	0.12	-0.25	0.14	-2.86	-0.37	-0.53	-1.25	-0.83	1.85	-0.03	0.70	0.69
T	-0.87	2.11	1.28	0.43	-2.89	1.22	-0.77	1.24	0.85	-0.47	-1.44	-0.53	-0.64	0.24	0.61	0.25	1.36
Motif	G	A	A	C	A	A	T	A	G	C	A	G	A	A	G	A	A

b) RAP1

Position	1	2	3	4	5	6	7	8	9	10	11
A	-2.21	0.30	3.78	-2.59	1.25	-0.73	3.44	-1.09	0.61	0.44	-2.16
C	-0.18	0.04	3.85	-2.73	0.13	-0.88	-12.28	-1.53	0.98	-0.81	-2.30
G	-1.77	-0.05	3.96	2.78	2.51	4.27	3.64	-1.20	0.02	1.37	0.25
T	-1.70	-2.43	4.62	-2.76	1.30	-0.98	-6.44	-1.00	0.56	-1.13	-1.62
Motif	A	T	A	T	A	T	C	C	A	T	C

c) GCN4

Position	1	2	3	4	5	6	7	8	9	10
A	-2.73	-0.09	2.932	-2.73	-4.26	-1.82	-1.68	1.118	1.892	-0.27
C	-0.93	0.35	-0.07	-0.21	-3.93	0.809	1.721	0.516	-1.61	1.279
G	-0.84	0.645	-0.49	1.094	-2.35	-0.81	1.101	-1.69	2.034	0.501
T	1.924	-1.57	-0.95	-0.83	-3.15	-1.13	2.073	3.276	2.806	0.701
Motif	A	T	T	A	A	A	A	G	C	A

Figure 1: EBMF-found Motif Results

5 DISCUSSION

In this paper, we have introduced an energy-based motif finding algorithm based on Francis *et al.*'s previous work. Unlike most of the currently available motif finding algorithms, it can utilize weak-signal information from ChIP-on-Chip experiment data, thus greatly improve the performance of overall motif finding process. Our primary goal up to now has been implementing the basic EBMF algorithm in our own program.

The implemented EBMF algorithms was applied to the yeast ChIP-on-Chip data set with three transcription factors; GAL4, GCN4, and RAP1. Unlike our expectation, the implemented program could not correctly detect the known TFBS motifs

at all.

In order to make sure that our program is capable of solving linear equations correctly, we generated a random but non-singular matrixes of many different sizes. Our program could solve these linear equations and the results were consistent with those calculated by MATLAB (data not shown). However, when we applied the actual matrixes to the MATLAB, even the MATLAB could not solve them and returned infinity values with warnings of singularity up to the precision of machine. This implies that the problem lies in the matrix itself not the solving methods.

To overcome this singularity problem, we tried to modify the target matrixes by adding some very small non-zero real numbers to all zero elements in the matrix. This was expected to make the matrix less singular but still this method could not overcome the singularity problem.

Unlike the QR-decomposition methods, the Downhill Simplex Method, one of the least square methods, could give somewhat reasonable energy values in Figure 1. However, this method could not find appropriate TFBS motifs, either. This might be because the Downhill Simplex Method is only a approximate solution and it may be caught in the local minimum at certain stages.

The goal of this project is to develop an improved EBMF algorithm that can utilize association information between positions in a motif, whose length may not be known. However, due to the unexpected singularity problems, we can not further move on to apply the distant relations between TFBS positions. Once the singularity problem is solved, we will be able to incorporate distant-relations into our EBMF algorithms along with the following improvements.

- Implementation of either the Powell's method[11] or the conjugate gradient method[11] for faster linear equation solution than the currently implemented downhill simplex method.
- Improvement of the initial energy matrix generation that allows various lengths of target motifs.

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